

The Perturbation Catalogue

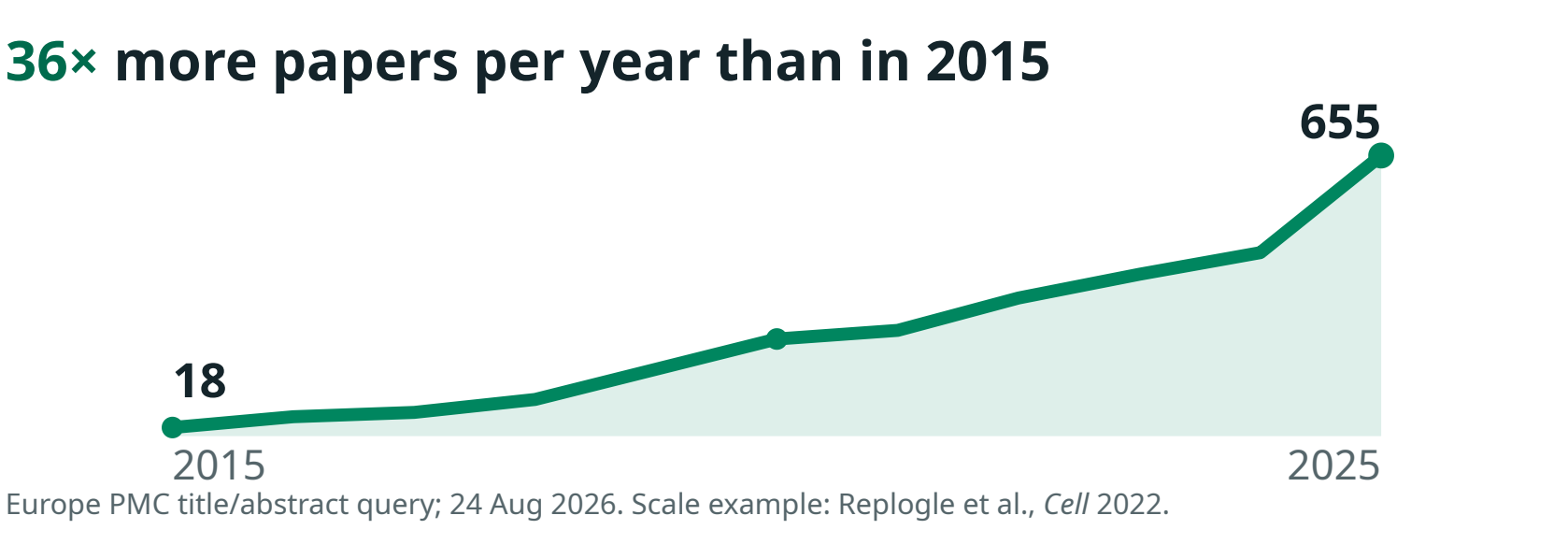
An Integrated Resource for Exploring the Impact of Genetic Perturbations

Kirill Tsukanov · Aleksandr Zakirov · Alexey Sokolov ·
Mallory Freeberg
EMBL European Bioinformatics Institute, Wellcome
Genome Campus, Hinxton, United Kingdom

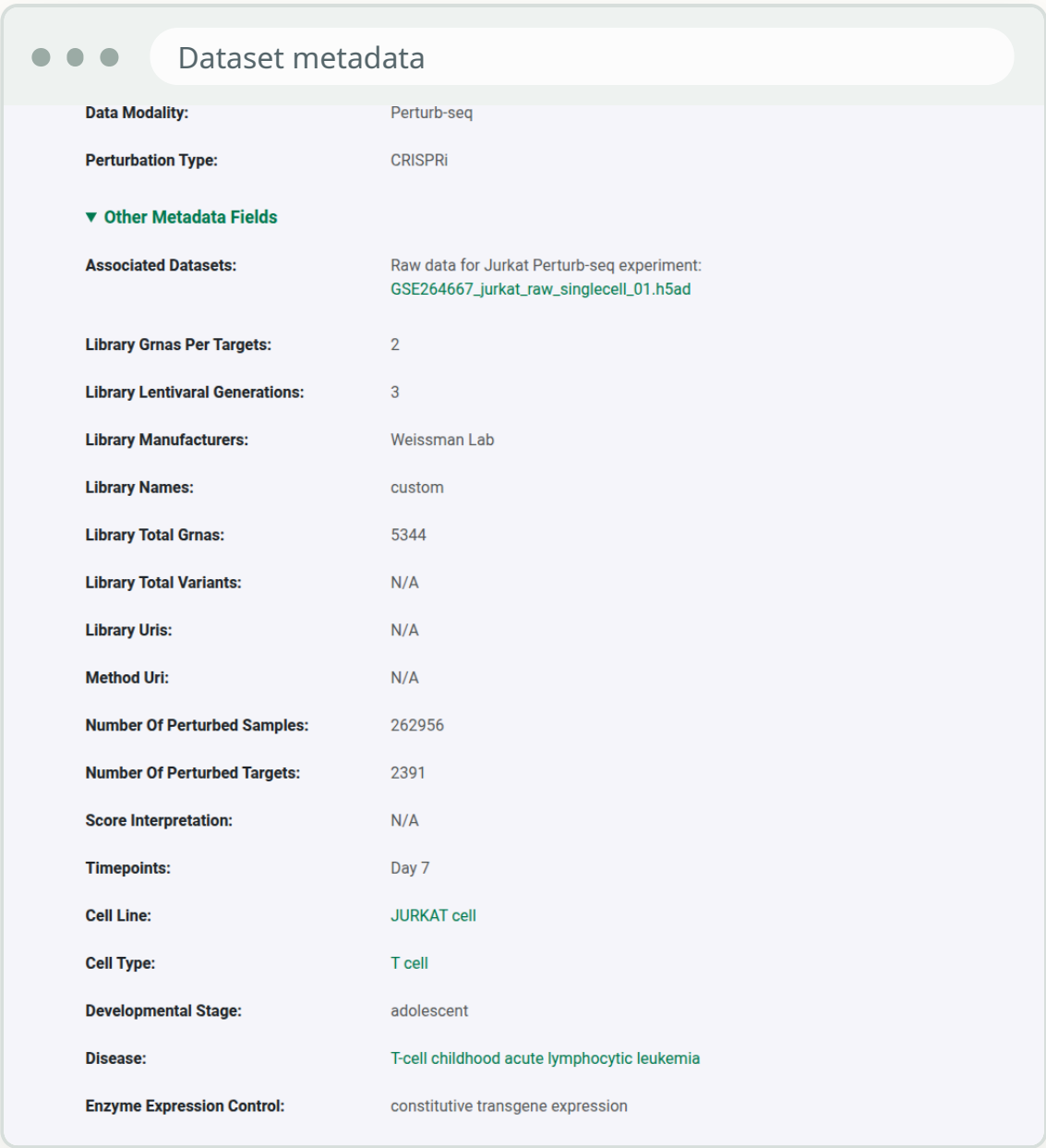
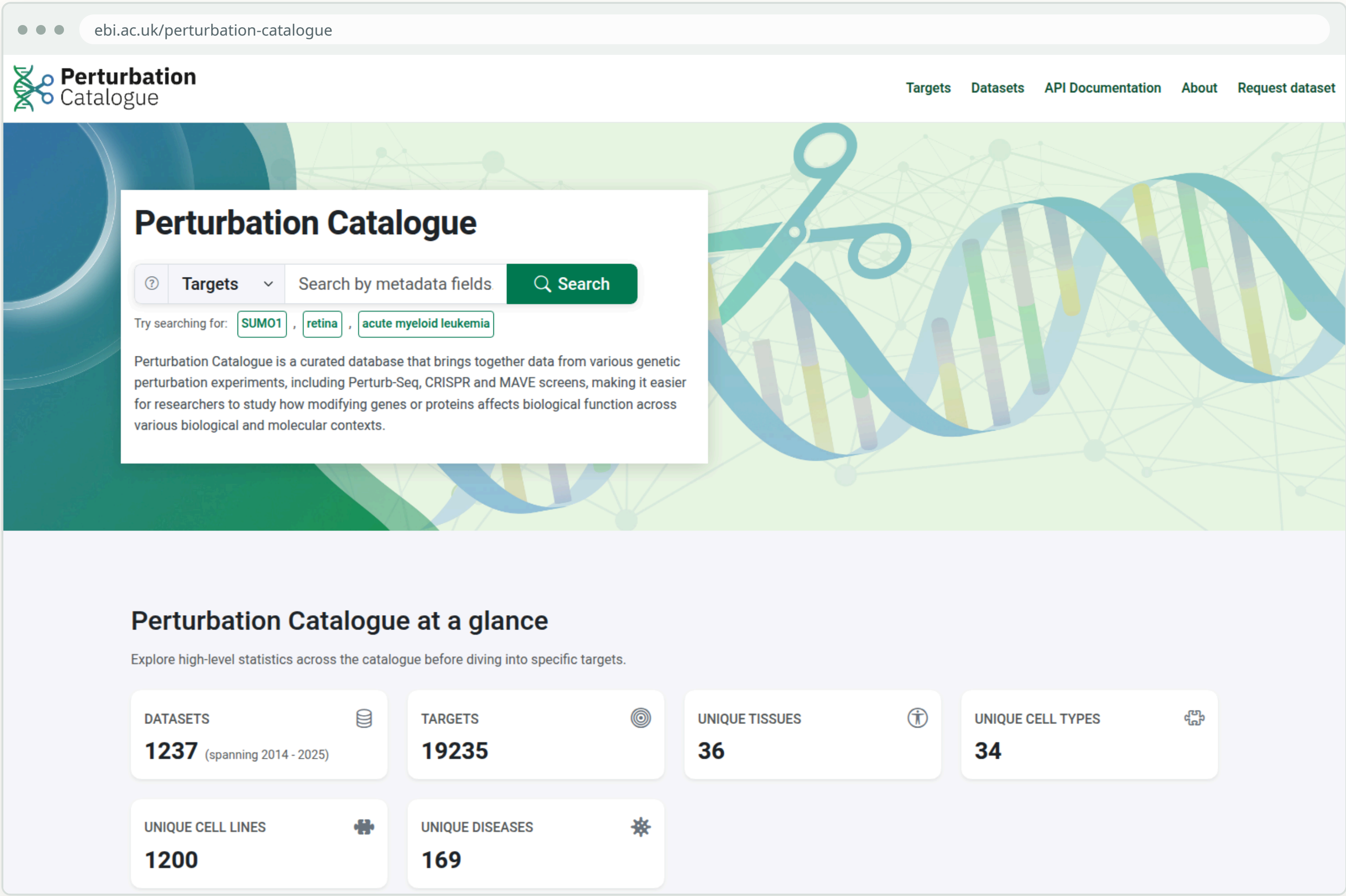
Perturbation evidence is multiplying faster than it can be organised

One genome-scale Perturb-Seq study profiled **more than 2.5 million cells**—but reuse still begins by reconciling incompatible studies.

Repositories · identifiers · metadata depth · score semantics · processing choices



The Catalogue makes studies interoperable without detaching results from context



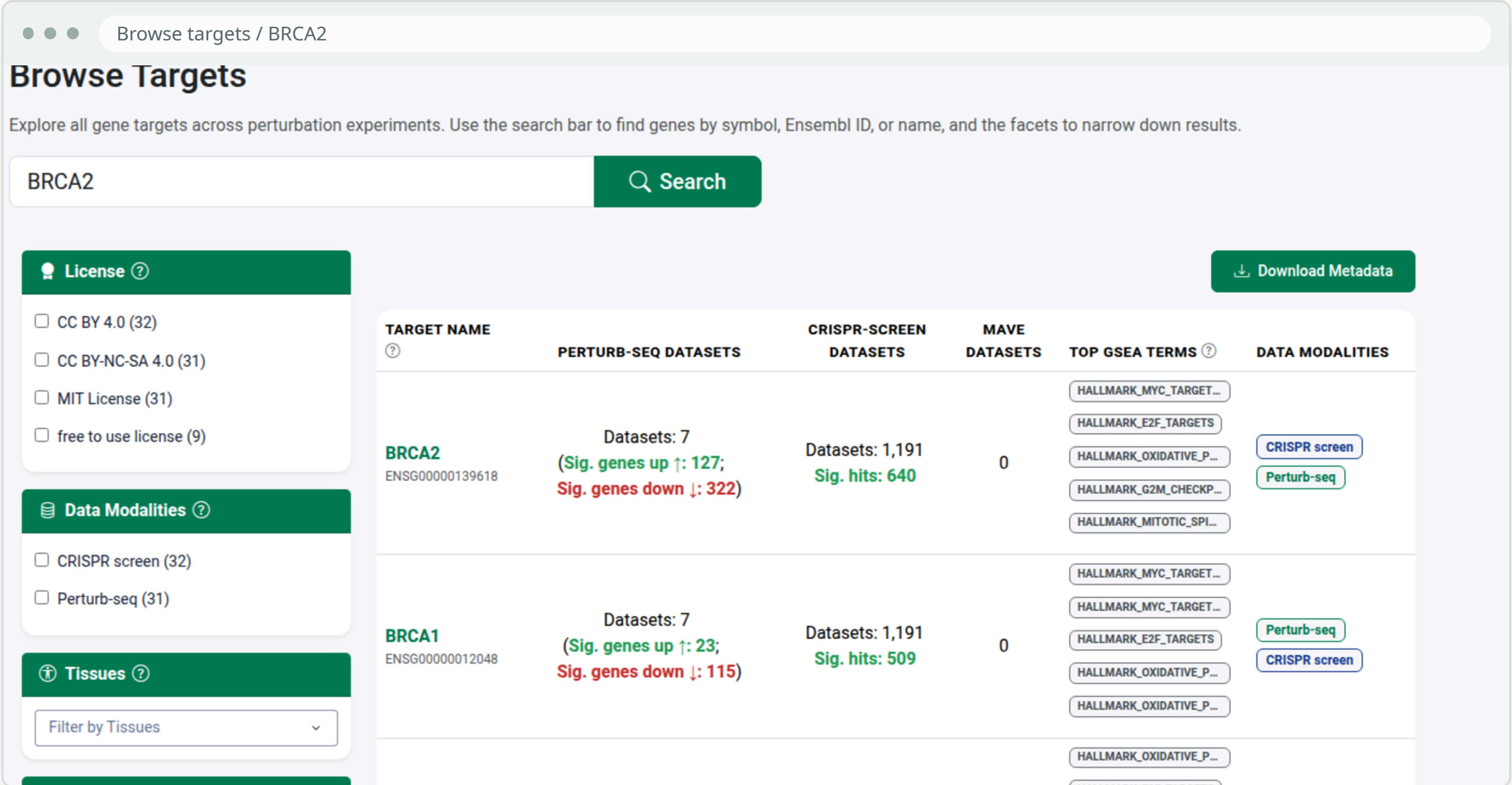
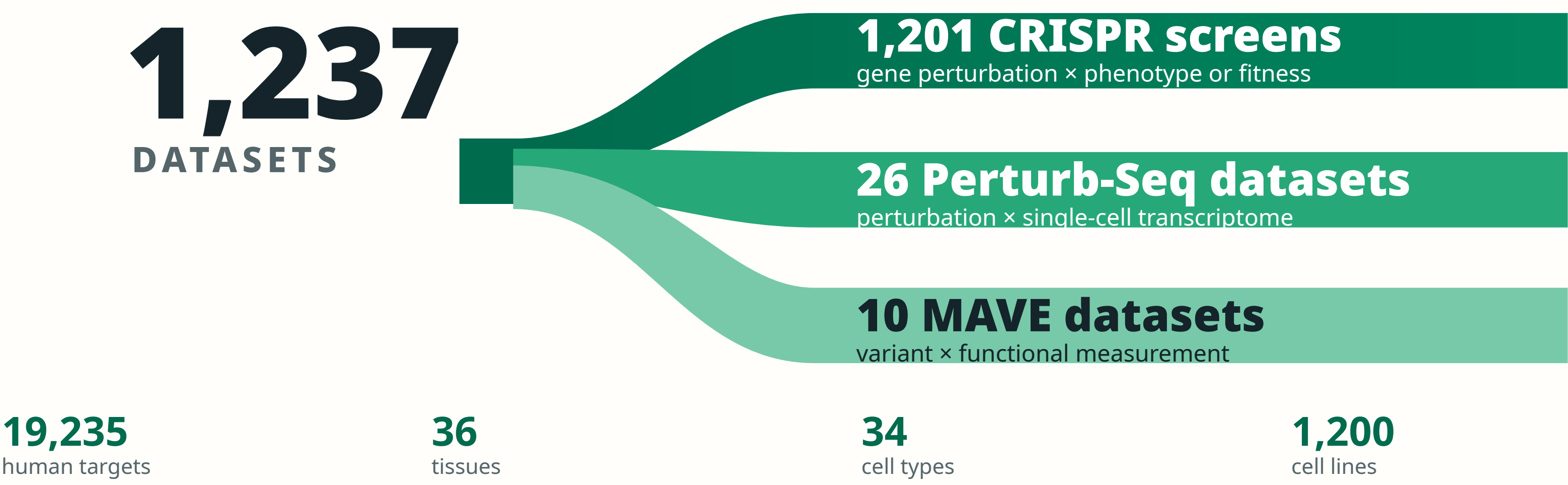
Metadata remains attached to every dataset

A validated, extensible schema records the experiment needed to interpret a score—not merely its accession.

Study · perturbation · assay · model system · tissue · cell type or line · disease · timepoint · licence · provenance

Ensembl gene IDs provide stable target identity; biological entities link to established ontologies. Full metadata is downloadable as JSON.

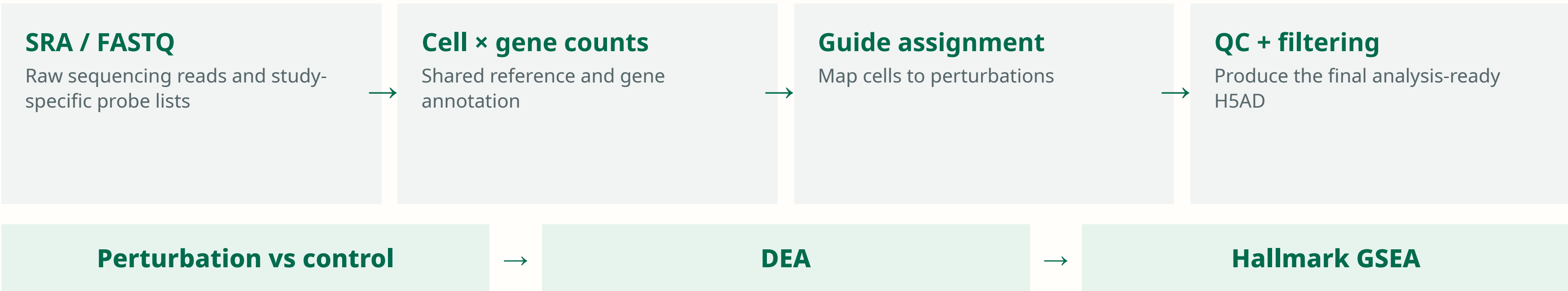
One Catalogue already connects 1,237 datasets across three complementary modalities



The BRCA2 view connects contributing datasets, significant CRISPR hits, Perturb-Seq effects and enriched Hallmark pathways. Production snapshot: 24 August 2026.

Perturb-Seq reprocessing starts again from raw reads

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.

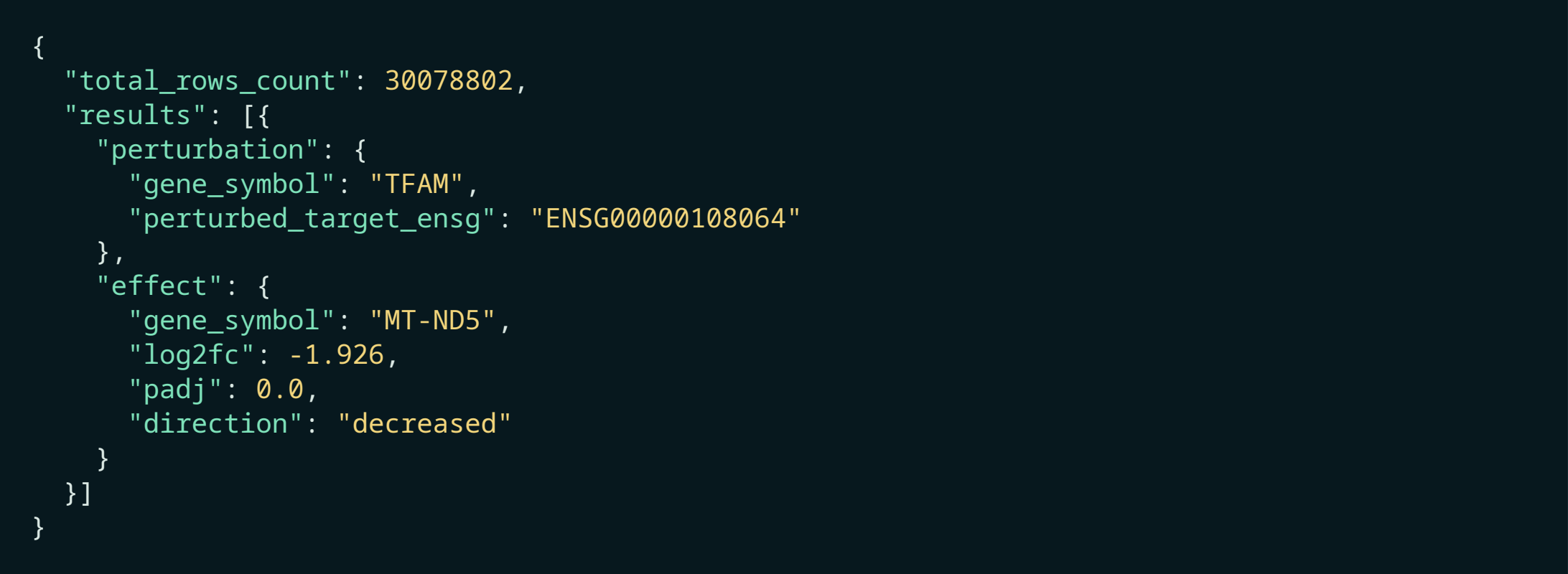


Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```



Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

perturbation-catalogue-help@ebi.ac.uk

Explore the Catalogue



EMBL-EBI

Open Targets